

## **Week 1: Project Kickoff and Planning (30th October - 5th November)**

### **- Tasks:**

- Kickoff meeting with Eviden team to discuss project goals and expectations.
- Finalise use case selection:
  - Being worked on by Jacob and Humais - Should have 2-3 each
- Set up project communication channels and version control.
  - Suggest in the next meeting we use Trello - alongside Whatsapp and Teams
- Create an initial Pitch deck containing current market solutions - how ours will stand out.
  - Martin is tasked to create the pitch
  - Aaron, Dixon and Daniel are tasked to research this and send their findings to Martin
- Set up recurring meetings with the company contact.
  - Martin is tasked to do this
- Creating a detailed project plan, outlining what needs to be done week by week, defining tasks and milestones - Andreas has been tasked to do this - May create a Gantt Chart as well

### **- Deliverables:**

- Use Case selection - can compile all of them into one document
- Powerpoint for Pitch
- Project Plan with detailed tasks and timelines

## **Week 2: Research and Design (6th November - 12th October)**

### **- Tasks:**

- Conduct in-depth research on chosen use cases
  - Company Contact may send us data to use - if so, create use cases based on them
  - Try and find large amounts of data we can supply our AI with
  - Make sure this process is documented thoroughly - requirements gathering
- Create an in-depth data flow diagram/Class Diagram of how the prototype and the final system should look
  - Give us an idea on what components we can implement now/leave for the final product.
  - What we need to actually implement/design - Document this thoroughly
- Create an Object diagram showing the relationship between different components of the project.
  - Use this for the database.
- Design system architecture.
- Begin designing the user interface/dashboards - Log-in System?
  - Company Contact said they would send some info on branding etc
  - Andreas can invite people to his figma project to design the initial UI design.
    - <https://www.figma.com>
  - Using Figma will allow us to show what the UI will look like and its functionality.

### **- Deliverables:**

- Research findings report for Use Cases.
- Detailed Data flow diagram.
- Initial UI/Dashboard design (Potentially along with a demo video) - Ask for feedback

### **Week 3: Development Phase 1 (13th November - 19th November)**

#### **- Tasks:**

- Assign team members to different components of the project
  - Design & Functionality of User Interface/Dashboard - Andreas, Aaron
  - Database Management - Humais, Dixon
  - Initial AI development - Jacob, Daniel, Mateusz
  - Communications and Pitches - Martin
- Set up the development environment.
  - Using PyCharm for the AI - make sure that PyCharm works in your chosen IDE.
  - Make sure that our Git repository is set up properly.
  - Make sure everyone has their own branches for when they code on Git.
    - As Git Admin, Mateusz will have to set this up.
  - Assign jobs and deliverables are added to the Git Board.
- Implement backend infrastructure (database, server setup).
  - Choose a database management system (MySQL) based on your project requirements.
    - Advantages of using the MySQL system:
      - Lightweight and serverless, making it easy to set up and use.
      - No configuration or maintenance required.
      - Suitable for small to medium-sized applications.
    - Design the database schema considering the data points you need to store for the chosen use case (Will depend on the use cases we pick).
    - Implement database tables, relationships, and define necessary indexes for optimised queries.
    - Develop user interfaces and initial dashboard prototypes.
      - Implement feedback on UI given by the company contact.
    - Start integrating AI technologies for data analysis.

#### **- Deliverables:**

- Backend infrastructure and database setup.
- Improved user interfaces and dashboard prototypes.

### **Week 4: Development Phase 2 (20th November - 26th November)**

#### **- Tasks:**

- Frontend Development:
  - Use frontend technologies like HTML, CSS, and JavaScript frameworks (e.g., React, Angular) to create interactive user interfaces.
  - Design intuitive and user-friendly interfaces for accessing the application and interacting with the data.
  - Develop initial dashboard prototypes showcasing essential features and data visualisations related to the selected use case (e.g., real-time energy consumption graphs, environmental impact metrics).
- User Experience (UX) Design:
  - Focus on creating a seamless user experience, ensuring consistency in design elements and navigation.
  - Gather feedback from team members and stakeholders to refine the user interface designs iteratively.

- Continue developing AI components for predictive modelling.
  - Utilise scikit-learn for building AI models. Will use tools like Anaconda to create isolated Python environments.
  - Clean, preprocess, and transform the data for training the AI models. Use tools like Pandas for data manipulation and analysis. Ensure the dataset is representative of the problem domain and includes relevant features for accurate predictions.
  - Choose appropriate machine learning algorithms based on the nature of the predictive modelling task. For instance, regression algorithms like Linear Regression or complex algorithms like Neural Networks (Deep Learning) can be selected based on the complexity of the problem

**- Deliverables:**

- Evidence of AI-driven features being worked on
- Evidence of the start of UI/ Dashboard design and functionality implementation process

**Week 5: Testing and Optimization (27th November - 3rd December)**

**- Tasks:**

- Implement security measures and access control systems.
- Integrate AI-driven features into the user interfaces - If possible.
  - Make sure everything is being documented.
- Conduct initial testing and debugging of UI and dashboard.
  - Try to finish the implementation of the UI.
  - Prepare documentation for end-users - how to use the UI.

**- Deliverables:**

- Testing reports and User Manuals for the UI - Potential Demo video
- Documentation for AI-driven features

**Week 6: Finalisation and Prototype Release (4th December - 10th December)**

**- Tasks:**

- Finalisation of the UI:
  - Make sure that it is usable and contains all the critical features needed for the prototype.
  - Conduct extensive testing, including functional, performance, and security testing.
- Optimise codebase and improve system performance.
- Address any remaining issues and bugs identified during testing.
- Prepare for the prototype demonstration.
- Present the Proof of Concept (POC) prototype to the Eviden team.
- Finalise documentation, including system architecture and codebase.

**- Deliverables:**

- Finalised prototype ready for demonstration.
- Presentation materials for the Eviden team.

**Week 7: Final Tasks: (11th December - 14th December)**

**- Task:**

- Make sure every team member has done the interim and individual report for the 15th of December.